

Project Proposal: Adaptive Manipulability-Based Path Planning for Robotic Arm Manipulators

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Project Title:

Implementing Adaptive Manipulability-Based RRT* for Optimized Robotic Arm Path Planning

Background and Motivation

Robotic manipulators are essential in industrial applications requiring high precision and reliability. Traditional motion planning methods, such as RRT and RRT, aim for feasible paths but often fail to optimize critical performance factors like manipulability, which determines the robot's dexterity and ability to avoid singularities. The paper "**Adaptive Manipulability-Based Path Planning Strategy for Industrial Robot Manipulators**" introduces an enhancement to RRT by integrating manipulability metrics into the path planning process. This ensures paths are not only short but also singularity-free and highly efficient for industrial robots.

Given my research interest in robotic arms for tasks such as arc welding and 3D printing, this approach aligns perfectly with my objectives to enhance trajectory optimization and robot dexterity in complex 3D environments.

Scope and Contributions of the Paper

The proposed Adaptive Manipulability-Based RRT* combines:

- Manipulability-Aware Planning:**
 - Incorporates a bi-objective cost function that balances path length and manipulability, ensuring singularity-free motion.
- Adaptive Step Size:**
 - Adjusts the RRT* tree's growth step dynamically based on manipulability and proximity to obstacles or goals, improving search efficiency.
- Constrained Closed-Loop Inverse Kinematics:**
 - Resolves redundancy in joint configurations to maximize manipulability while ensuring precise end-effector positioning.

The methodology has been validated in simulation and real-world experiments using a 6-DOF FANUC manipulator, demonstrating improved trajectory tracking and manipulability.

Project Scope

- Objective:**
 - Implement the Adaptive Manipulability-Based RRT* algorithm for a robotic manipulator (e.g., Panda or Tiago arm) using ROS2.
 - Focus on optimizing motion for tasks involving welding or 3D printing on complex surfaces.
- Implementation:**
 - Develop a ROS2-compatible implementation of Adaptive Manipulability-Based RRT*.
 - Use MoveIt! for collision checking and trajectory execution, and leverage tools like Gazebo for simulation.
- Testing:**
 - Evaluate the algorithm in simulated environments with varying workspace complexities.
 - Compare results with standard RRT* to assess improvements in path quality, manipulability, and computational efficiency.
- Potential Extensions:**
 - Incorporate real-time updates to adapt to dynamic environments.
 - Explore hybrid approaches by integrating learning-based sampling distributions for further optimization.

Expected Outcomes

- An implementation of Adaptive Manipulability-Based RRT* for robotic arm manipulators.
- Detailed performance evaluation demonstrating the algorithm's effectiveness in maintaining high manipulability while optimizing path length.
- Insights into the feasibility and scalability of this approach for industrial and research applications, particularly for tasks requiring precise manipulator trajectories.

This project will provide a practical understanding of manipulability-aware path planning and its integration into modern robotic systems, forming a foundation for further exploration of advanced motion planning techniques in dynamic and complex environments.

Reference

H. Shen, W. -F. Xie, J. Tang and T. Zhou, "Adaptive Manipulability-Based Path Planning Strategy for Industrial Robot Manipulators," in IEEE/ASME Transactions on Mechatronics, vol. 28, no. 3, pp. 1742-1753, June 2023, doi: 10.1109/TMECH.2022.3231467. keywords: {Robots;Cost function;Kinematics;Service robots;End effectors;Trajectory;Jacobian matrices;Adaptive step size;manipulability;path planning;rapidly-exploring random tree (RRT*);robot manipulator},